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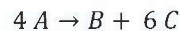
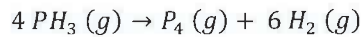
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CHEMICAL REACTION ENGINEERING (CHE 321)

MIDTERM EXAM 1

CLOSED BOOK, TIME 70 MIN

The homogeneous gas-phase decomposition of phosphine



proceeds at 922.15 K with the first-order rate:

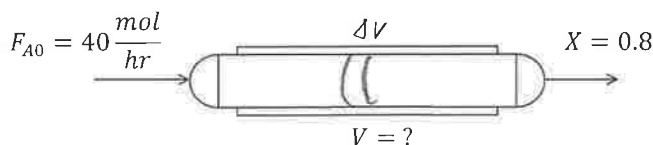
$$r_A = -k C_A$$

$$k = 10 \text{ hr}^{-1}$$

- 35 1. What size of plug flow reactor (PFR), operating at constant temperature and pressure of 922.15 K and 460 kPa, can achieve 80% conversion of a feed of 40 moles of pure phosphine per hour?

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$. Gas constant $R = 8.314 \frac{\text{Pa} \cdot \text{m}^3}{\text{mol} \cdot \text{K}}$. $\int dx \frac{1+\epsilon x}{1-x} = -(1+\epsilon) \ln(1-x) - \epsilon x$.

- 5 1.1. Based on a shell mole balance, derive a differential equation for conversion as a function of PFR volume.
15 1.2. Express the rate of reaction in terms of conversion X .
15 1.3. Integrate the differential mole balance and determine the volume.



$$r_A = -k C_A$$

$$pV = nRT$$

$$X = \frac{F_{A0} - F_A}{F_{A0}}$$

$$\frac{V_0}{n_0} = \frac{V}{n}$$

$$C_{A0} = \frac{F_{A0}}{V_0}$$

$$v_0 n = v n_0$$

$$\frac{V}{v_0} = \frac{n}{n_0}$$

$$C_A = \frac{F_{A0} - F_{A0}X}{v}$$

$$\frac{v}{v_0} = \left(\frac{1+\epsilon x}{1-x} \right)$$

$$-r_A = k \frac{F_{A0} - F_{A0}X}{\left(\frac{1+\epsilon x}{1-x} \right) \cdot V_0}$$

$$\frac{dX}{dV} = k \frac{F_{A0} - F_{A0}X}{\left(\frac{1+\epsilon x}{1-x} \right) \cdot V_0}$$

$$\frac{T_0}{T} = 1$$

$$\frac{p}{p_0} = 1$$

gas phase

$$\frac{F_T}{F_{T0}} \downarrow$$

$$\frac{F_T}{F_{T0}} = \frac{1-x}{1+\epsilon x}$$

$$\epsilon = (7-4)(1)$$

$$\boxed{\epsilon = 3}$$

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$$1) \frac{dx}{dV} = K F_{A0} - F_{A0} x / \left(\left(\frac{1+\epsilon x}{1-x} \right) V_0 \right)$$

$$\int dx \cdot \left(\frac{1+\epsilon x}{1-x} \right) V_0 + F_{A0} x = \int dV K F_{A0}$$

$$-(1+\epsilon) \ln(1-x) - \epsilon x + \frac{F_{A0} x^2}{2} = V K F_{A0}$$

$$\frac{-(1+\epsilon) \ln(1-x) - \epsilon x + \frac{F_{A0} x^2}{2}}{K F_{A0}} = V$$

$$\epsilon = 3$$

$$x = 0.8$$

$$F_{A0} = 40 \frac{\text{mol}}{\text{hr}}$$

$$K = 10 \frac{1}{\text{hr}}$$

$$\frac{-(4) \ln(0.2) - 2.4 + \frac{40(0.8)^2}{2}}{10 \cdot 40}$$

$$\frac{-3.22 - 2.4 + 12.8}{400}$$

$$= 0.0421 \text{ m}^3$$

$$42.1 \text{ dm}^3$$

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$$-1268 = -(1+\epsilon) \ln(1-x) - \epsilon x$$

$$\epsilon \approx .176$$

x too big for ϵ

$$-1268 = -(1.176) \ln(1-x) - \epsilon x$$

$$\frac{-1268 + .176x}{-1.176} = \ln(1-x)$$

$$e^{10.78} + e^{.176x} = 1-x$$

$$2.94 + e^{.176x} = 1-x$$

$$1.94 + e^{.176x} = -x$$

$$1.94 = -x - e^{.176x}$$